Keldysh and Mori–Zwanzig

Two languages for the same open-system dynamics:
a working note and a self-study plan

Version 2

Notes prepared with Claude

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About the references. All bibliography entries were checked on 30 September 2026 against publisher, Crossref, or arXiv records, and DOIs were added where they exist. The equations in the main text were derived and cross-checked independently of the references.

Changes in v2. Section 2 was rewritten and expanded: it now derives the contour from the second-order term of $U\rho U^\dagger$, works out the Keldysh rotation on a qubit, derives the 2×2 (Larkin–Ovchinnikov) structure of the bath contractions from the $\pm$ basis, checks the second-order and fourth-order diagrams explicitly, and derives the influence functional as the commutative limit of the Wick sum. Equation numbers after Section 2 have shifted; labels are unchanged.

The thesis in three lines

1. *Density-matrix level.* Ordering Liouville-space superoperators along a single real time axis *is* Keldysh contour ordering. With the Nakajima projector $\mathcal{P} = \rho_B \otimes \mathrm{Tr}_B$ and a Gaussian bath, the NZ memory kernel is the *irreducible self-energy* of the reduced propagator, and the NZ equation is a Dyson equation.
2. *Operator level.* Mori's projection onto a system operator A yields a generalized Langevin equation. For $A = c$ with the anticommutator inner product, the Mori memory function is (up to a factor $-i$) the *retarded* self-energy Σ^R.
3. *Common ground.* Both expansions are built from the two bath kernels that appear in the Keldysh/Feynman–Vernon influence functional: the response χ (retarded, c–q) and the noise S (Keldysh, c–c). Differences between the methods come from *which object is projected* and *how the series is resummed*, not from different physics.

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1 Setup and conventions

We set $\hbar = 1$ throughout and take

$$H = H_S + H_B + H_I, \qquad H_I = \sum_\alpha S_\alpha \otimes B_\alpha, \tag{1}$$

with Hermitian S_α, B_α, a stationary bath state ρ_B ($[H_B, \rho_B] = 0$), and $\langle B_\alpha \rangle \equiv \mathrm{Tr}_B(B_\alpha \rho_B) = 0$. The Liouvillian is $\mathcal{L}X = -i[H, X] = (\mathcal{L}_S + \mathcal{L}_B + \mathcal{L}_I)X$. For a factorized initial state $\rho(0) = \rho_S(0) \otimes \rho_B$, the reduced dynamics defines the dynamical map

$$\rho_S(t) = \Pi(t)\, \rho_S(0), \qquad \Pi(t)X \equiv \mathrm{Tr}_B\big[\mathrm{e}^{\mathcal{L}t}(X \otimes \rho_B)\big]. \tag{2}$$

Left/right superoperators and the Keldysh rotation. For any operator A, define $A^+X = AX$ (acting on the ket, i.e. the *forward* branch) and $A^-X = XA$ (acting on the bra, i.e. the *backward* branch). Then $\mathcal{L}_I = -i(H_I^+ - H_I^-)$. The Keldysh rotation to "classical" and "quantum" combinations is

$$A^{\mathrm{c}} = \tfrac{1}{2}\big(A^+ + A^-\big) \ \ \big(A^{\mathrm{c}}X = \tfrac{1}{2}\{A, X\}\big), \qquad A^{\mathrm{q}} = A^+ - A^- \ \ \big(A^{\mathrm{q}}X = [A, X]\big). \tag{3}$$

Since system and bath superoperators commute, $S^+B^+ - S^-B^- = S^{\mathrm{c}}B^{\mathrm{q}} + S^{\mathrm{q}}B^{\mathrm{c}}$, and therefore

$$\boxed{\mathcal{L}_I = -i \sum_\alpha \big(S_\alpha^{\mathrm{q}} B_\alpha^{\mathrm{c}} + S_\alpha^{\mathrm{c}} B_\alpha^{\mathrm{q}}\big).} \tag{4}$$

Two trace identities do most of the work: $\mathrm{Tr}_B(B^{\mathrm{q}}Y) = 0$ for any Y (the trace of a commutator vanishes), and $\mathrm{Tr}_B(B^{\mathrm{c}}Y) = \mathrm{Tr}_B(BY)$.

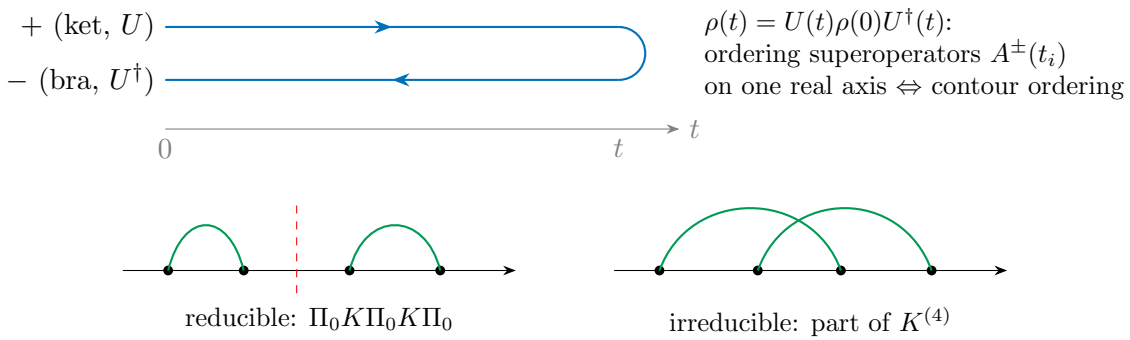


Figure 1: Top: the Keldysh contour. The forward and backward branches are the ket and bra sides of ρ. Bottom: real-time diagrams for the reduced propagator. Vertices are system superoperators $S^{\mathrm{c/q}}$ and lines are bath contractions (S or χ). A diagram that can be cut between vertices without cutting a line (red) is reducible. The NZ kernel collects exactly the irreducible diagrams.

Bath kernels. With $B_\alpha(t) = \mathrm{e}^{iH_B t} B_\alpha \mathrm{e}^{-iH_B t}$,

$$S_{\alpha\beta}(t) = \tfrac{1}{2}\langle\{B_\alpha(t), B_\beta(0)\}\rangle \qquad \text{(symmetrized noise; Keldysh component)}, \tag{5}$$

$$\chi_{\alpha\beta}(t) = i\,\theta(t)\langle[B_\alpha(t), B_\beta(0)]\rangle \qquad \text{(Kubo response; retarded component)}. \tag{6}$$

The full correlator is $C_{\alpha\beta}(t) = \langle B_\alpha(t) B_\beta\rangle = S_{\alpha\beta}(t) - \frac{i}{2}\chi_{\alpha\beta}(t)$ for $t > 0$. For a single Hermitian B, $S(t)$ is real and even. In thermal equilibrium the fluctuation–dissipation theorem (FDT) reads $S(\omega) = \coth(\beta\omega/2)\,\chi''(\omega)$, with $S(\omega) = \int \mathrm{d}t\, \mathrm{e}^{i\omega t} S(t)$ and $\chi'' = \mathrm{Im}\,\chi$. Up to factors of i, these are the bath Green's functions D^K and D^R of Keldysh field theory [1, 2].

2 Keldysh on the density matrix: why the contour is already there

The claim of this section is that nothing has to be *added* to the von Neumann equation to obtain the Keldysh contour. The contour, the c/q rotation, the 2×2 matrix structure of the bath propagator and the causality rules all appear on their own once one treats ρ as a vector and H as a pair of superoperators. We go through this slowly, with one worked example for each step. The results that the rest of the note depends on are Eq. (22) (the contractions) and Eq. (26) (the influence functional); the intermediate steps are there to make those two formulas feel inevitable.

2.1 The contour from the von Neumann equation

Work in the interaction picture with respect to $H_0 = H_S + H_B$, so that $\tilde{H}_I(t) = \mathrm{e}^{iH_0 t} H_I \mathrm{e}^{-iH_0 t}$ and $\tilde{\rho}(t) = \tilde{U}(t)\rho(0)\tilde{U}^\dagger(t)$ with

$$\tilde{U}(t) = T \exp\Big[-i\int_0^t \mathrm{d}s\, \tilde{H}_I(s)\Big] = 1 - i\int_0^t \mathrm{d}t_1 \tilde{H}_I(t_1) - \int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2\, \tilde{H}_I(t_1)\tilde{H}_I(t_2) + \dots \tag{7}$$

Its adjoint is *anti*-time-ordered: in $\tilde{U}^\dagger$ the later operator stands to the right, $\tilde{U}^\dagger = 1 + i\int \tilde{H}_I - \int_{t_1>t_2} \tilde{H}_I(t_2)\tilde{H}_I(t_1) + \dots$

Example: the second-order term, written out

Collecting all terms of order H_I^2 in $\tilde{U}\rho(0)\tilde{U}^\dagger$,

$$\begin{aligned}\tilde{\rho}^{(2)}(t) = &-\int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2\, \tilde{H}_I(t_1)\tilde{H}_I(t_2)\,\rho(0) && \text{(both on the ket side; time-ordered)}\\ &-\int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2\, \rho(0)\,\tilde{H}_I(t_2)\tilde{H}_I(t_1) && \text{(both on the bra side; anti-time-ordered)}\\ &+\int_0^t \mathrm{d}t_1 \int_0^{t} \mathrm{d}t_2\, \tilde{H}_I(t_1)\,\rho(0)\,\tilde{H}_I(t_2) && \text{(one on each side; } \textit{no} \text{ mutual ordering).}\end{aligned} \tag{8}$$

The third term is the one that looks unusual: t_1 and t_2 are integrated independently over $[0,t]$, so there is no time ordering between a ket-side and a bra-side operator. All three terms become a *single* ordering rule once we say that every point on the forward (+) branch is "earlier" than every point on the backward (−) branch, that time runs forward on + and backward on −, and that operators are ordered with later contour positions to the *left* of $\rho(0)$ or to the *right* of it according to their branch. That rule is contour ordering, Fig. 1 (top) [3, 4].

The superoperator language makes the rule automatic. With $A^+X = AX$ and $A^-X = XA$,

$$\tilde{\rho}^{(2)}(t) = \int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2\, \tilde{\mathcal{L}}_I(t_1)\tilde{\mathcal{L}}_I(t_2)\,\rho(0), \qquad \tilde{\mathcal{L}}_I(s) = -i\big[\tilde{H}_I^+(s) - \tilde{H}_I^-(s)\big], \tag{9}$$

and multiplying out the two brackets reproduces the three terms of Eq. (8): H^+H^+ gives the first, H^-H^- gives the second (note $H^-(t_1)H^-(t_2)\rho = \rho H(t_2)H(t_1)$, automatically anti-ordered), and the cross terms $H^+(t_1)H^-(t_2) + H^-(t_1)H^+(t_2)$ give $H(t_1)\rho H(t_2) + H(t_2)\rho H(t_1)$ restricted to $t_1 > t_2$, which is the third term with its two orderings separated. In other words,

Statement

Ordinary time ordering of the superoperators $\tilde{\mathcal{L}}_I(t_i)$ along the single real axis $[0,t]$, with a sum over the branch index $\pm$ at each vertex, *is* Keldysh contour ordering. One never has to think about the contour again: it is encoded in the sign of $\mathcal{L}_I$ and the left/right action of $H^\pm$.

2.2 The Keldysh rotation in Liouville space

The $\pm$ basis is natural for deriving the contour but awkward for calculations, because the two branches are strongly correlated (the trace at the final time glues them together). The Keldysh rotation (3) separates "what the two branches have in common" (c) from "how they differ" (q). Inverting it,

$$A^+ = A^{\mathrm{c}} + \tfrac{1}{2}A^{\mathrm{q}}, \qquad A^- = A^{\mathrm{c}} - \tfrac{1}{2}A^{\mathrm{q}}. \tag{10}$$

Derivation of Eq. (4). Since S and B act on different factors of the Hilbert space, $S^\pm$ and $B^\pm$ commute and $(SB)^\pm = S^\pm B^\pm$. Then with Eq. (10),

$$\begin{aligned}S^+B^+ - S^-B^- &= \big(S^{\mathrm{c}} + \tfrac{1}{2}S^{\mathrm{q}}\big)\big(B^{\mathrm{c}} + \tfrac{1}{2}B^{\mathrm{q}}\big) - \big(S^{\mathrm{c}} - \tfrac{1}{2}S^{\mathrm{q}}\big)\big(B^{\mathrm{c}} - \tfrac{1}{2}B^{\mathrm{q}}\big)\\ &= S^{\mathrm{c}}B^{\mathrm{q}} + S^{\mathrm{q}}B^{\mathrm{c}},\end{aligned} \tag{11}$$

the $S^{\mathrm{c}}B^{\mathrm{c}}$ and $\frac{1}{4}S^{\mathrm{q}}B^{\mathrm{q}}$ terms cancelling. Hence $\mathcal{L}_I = -i\sum_\alpha (S^{\mathrm{q}}_\alpha B^{\mathrm{c}}_\alpha + S^{\mathrm{c}}_\alpha B^{\mathrm{q}}_\alpha)$, which is Eq. (4). The structure "one c, one q per vertex" is the same as in the Keldysh action, where the interaction term $H_+ - H_-$ is odd in the quantum field and so always contains at least one q leg.

What c and q do to a density matrix. It helps to see the two superoperators act on something concrete.

Example: σ_z^{c} and σ_z^{q} on a qubit

Write $\rho = \sum_{ij} \rho_{ij} |i\rangle\langle j|$ with $\sigma_z|i\rangle = z_i|i\rangle$, $z_0 = +1$, $z_1 = -1$. Then

$$\sigma_z^{\mathrm{c}} |i\rangle\langle j| = \tfrac{1}{2}\{\sigma_z, |i\rangle\langle j|\} = \tfrac{z_i+z_j}{2}|i\rangle\langle j|, \qquad \sigma_z^{\mathrm{q}} |i\rangle\langle j| = [\sigma_z, |i\rangle\langle j|] = (z_i - z_j)|i\rangle\langle j|. \tag{12}$$

In the Liouville basis $(|0\rangle\langle 0|, |0\rangle\langle 1|, |1\rangle\langle 0|, |1\rangle\langle 1|)$ both are diagonal:

$$\sigma_z^{\mathrm{c}} = \mathrm{diag}(1, 0, 0, -1), \qquad \sigma_z^{\mathrm{q}} = \mathrm{diag}(0, 2, -2, 0). \tag{13}$$

So σ_z^{c} reads out the populations and leaves coherences alone, while σ_z^{q} sees only the coherences and annihilates populations. This is the operator version of "the classical field is the physical configuration, the quantum field measures the difference between the two branches": a diagonal (classical-looking) density matrix has no q component with respect to σ_z.

Example: σ_x^{c} and σ_x^{q} on the same qubit

For σ_x the roles are mixed, since σ_x is off-diagonal in the z basis. Acting on the coherence $X = |0\rangle\langle 1|$,

$$\sigma_x^{\mathrm{c}} X = \tfrac{1}{2}\big(|1\rangle\langle 1| + |0\rangle\langle 0|\big) = \tfrac{1}{2}\mathbb{1}, \qquad \sigma_x^{\mathrm{q}} X = |1\rangle\langle 1| - |0\rangle\langle 0| = -\sigma_z. \tag{14}$$

The c action produces a traceful (population-like) object, the q action a traceless one. This is not a coincidence: $\mathrm{Tr}_S(A^{\mathrm{q}} X) = \mathrm{Tr}_S[A, X] = 0$ for every A and X, so A^{q} always maps into the traceless subspace. It is the reason the *last* vertex of any diagram must be S^{q} (see below): a final S^{c} would change $\mathrm{Tr}_S\, \rho_S$.

The two trace identities, proved. For an arbitrary operator Y on $S \otimes B$ and a bath operator B,

$$\mathrm{Tr}_B\big(B^{\mathrm{q}} Y\big) = \mathrm{Tr}_B\big(BY - YB\big) = 0, \qquad \mathrm{Tr}_B\big(B^{\mathrm{c}} Y\big) = \tfrac{1}{2}\,\mathrm{Tr}_B\big(BY + YB\big) = \mathrm{Tr}_B\big(BY\big), \tag{15}$$

both by cyclicity of the partial trace over the bath factor (which is allowed because B acts as the identity on S). The first identity is the superoperator form of the Keldysh statement that a correlation function with its *latest* point on a quantum leg vanishes, which is what makes $\langle \phi_{\mathrm{q}} \phi_{\mathrm{q}} \rangle = 0$ and makes $\langle \phi_{\mathrm{c}} \phi_{\mathrm{q}} \rangle$ retarded [1]. We will see it do exactly that in the next subsection.

2.3 Bath contractions and the 2×2 Keldysh structure

Consider a single Hermitian bath operator B (several couplings just add indices). The basic object is the time-ordered two-point function of bath superoperators,

$$D^{ab}(t, s) \equiv \mathrm{Tr}_B\big[T_{\mathcal{L}}\, \tilde{B}^a(t) \tilde{B}^b(s)\, \rho_B\big], \qquad a, b \in \{+, -\} \text{ or } \{\mathrm{c}, \mathrm{q}\}, \tag{16}$$

where $T_{\mathcal{L}}$ puts the later superoperator on the left. In the $\pm$ basis the four components are, by direct evaluation,

component	string ($t > s$)	value	
$D^{++}(t,s)$	$\mathrm{Tr}_B[B(t)B(s)\rho_B]$	$\langle T\, B(t)B(s)\rangle = C(t-s)$	(time-ordered)
$D^{--}(t,s)$	$\mathrm{Tr}_B[\rho_B B(s)B(t)]$	$\langle \tilde{T}\, B(t)B(s)\rangle = C(s-t)$	(anti-time-ordered)
$D^{+-}(t,s)$	$\mathrm{Tr}_B[B(t)\rho_B B(s)]$	$\langle B(s)B(t)\rangle = C(s-t)$	("lesser")
$D^{-+}(t,s)$	$\mathrm{Tr}_B[B(s)\rho_B B(t)]$	$\langle B(t)B(s)\rangle = C(t-s)$	("greater")

These are precisely the four contour-ordered Green's functions $D^T, D^{\tilde{T}}, D^<, D^>$ of Keldysh theory (up to the overall $-i$ in the usual definition), and the familiar relation $D^T + D^{\tilde{T}} = D^< + D^>$ holds term by term. Now rotate. Using $B^{\mathrm{c}} = \frac{1}{2}(B^+ + B^-)$ and $B^{\mathrm{q}} = B^+ - B^-$ on both legs:

$$D^{\mathrm{cc}}(t,s) = \tfrac{1}{4}\big(D^{++} + D^{+-} + D^{-+} + D^{--}\big) = \tfrac{1}{2}\big[C(t-s) + C(s-t)\big] = S(t-s), \tag{17}$$

$$D^{\mathrm{cq}}(t,s) = \tfrac{1}{2}\big(D^{++} - D^{+-} + D^{-+} - D^{--}\big) = \theta(t-s)\langle[B(t),B(s)]\rangle = -i\chi(t-s), \tag{18}$$

$$D^{\mathrm{qc}}(t,s) = \tfrac{1}{2}\big(D^{++} + D^{+-} - D^{-+} - D^{--}\big) = -\theta(s-t)\langle[B(t),B(s)]\rangle = -i\chi(s-t), \tag{19}$$

$$D^{\mathrm{qq}}(t,s) = D^{++} - D^{+-} - D^{-+} + D^{--} = 0. \tag{20}$$

(For the middle two lines, evaluate separately for $t > s$ and $t < s$ using the table; the θ functions appear because the "later" entry of the table switches.) In matrix form,

$$\boxed{D = \begin{pmatrix} D^{\mathrm{cc}} & D^{\mathrm{cq}} \\ D^{\mathrm{qc}} & D^{\mathrm{qq}} \end{pmatrix}(t,s) = \begin{pmatrix} S(t-s) & -i\chi(t-s) \\ -i\chi(s-t) & 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} D^K & D^R \\ D^A & 0 \end{pmatrix},} \tag{21}$$

the Larkin–Ovchinnikov structure [1, 5]. The zero in the qq slot and the retardation of the cq slot are both consequences of the first trace identity in Eq. (15): whenever the *later* superoperator in the string is B^{q}, the trace kills the term. Restricting to $t > s$ and reading off the entries gives Eq. (22):

$$\mathrm{Tr}_B[B^{\mathrm{c}}(t)B^{\mathrm{c}}(s)\rho_B] = S(t-s), \qquad \mathrm{Tr}_B[B^{\mathrm{c}}(t)B^{\mathrm{q}}(s)\rho_B] = -i\chi(t-s), \qquad \mathrm{Tr}_B[B^{\mathrm{q}}(t)\,\cdot\,] = 0. \tag{22}$$

Example: a single bath mode

Let $B = g(b + b^\dagger)$ with $H_B = \omega_b b^\dagger b$ and ρ_B thermal, $\bar{n} = (\mathrm{e}^{\beta\omega_b} - 1)^{-1}$. Then $C(t) = g^2[(\bar{n}+1)\mathrm{e}^{-i\omega_b t} + \bar{n}\,\mathrm{e}^{i\omega_b t}]$, and

$$S(t) = g^2(2\bar{n}+1)\cos\omega_b t, \qquad \chi(t) = 2g^2\,\theta(t)\sin\omega_b t. \tag{23}$$

The noise kernel carries the occupation and grows with temperature; the response kernel is temperature independent and is odd in t (before the θ), as a Kubo response must be. In frequency space, $\chi''(\omega) = \pi g^2[\delta(\omega-\omega_b) - \delta(\omega+\omega_b)]$ and $S(\omega) = \coth(\beta\omega/2)\,\chi''(\omega)$: the FDT, checked by hand. A continuum of modes with density $J(\omega) = \pi\sum_k g_k^2\delta(\omega-\omega_k)$ gives $\chi''(\omega) = J(\omega) - J(-\omega)$ and the usual spin-boson kernels [6].

Wick's theorem for superoperators. For a Gaussian ρ_B (free bosons or fermions in a thermal or any other Gaussian state), the contour-ordered n-point function of bath operators factorizes into a sum over all pairings of two-point functions. Because the Liouville string $\mathrm{Tr}_B[\tilde{B}^{a_n}(t_n)\cdots\tilde{B}^{a_1}(t_1)\rho_B]$ is literally a contour-ordered expectation value (Section 2.1), Wick's theorem holds for it unchanged, with each pair contributing $D^{a_i a_j}(t_i,t_j)$ from Eq. (21). For fermionic B a sign $(-1)^P$ of the pairing permutation appears; everything else is the same. Non-Gaussian baths replace pairings by cumulants [7].

2.4 System-side Feynman rules, and a check at second order

Insert Eq. (4) into the Dyson series for $\tilde{\rho}(t)$ and take Tr_B. Every vertex is either $S^{\mathrm{q}}B^{\mathrm{c}}$ or $S^{\mathrm{c}}B^{\mathrm{q}}$, so the bath index at a vertex is the *opposite* of the system index. Translating Eq. (21) to the system side:

System-side Feynman rules

Vertices are $-iS^{\mathrm{q}}(t)$ or $-iS^{\mathrm{c}}(t)$, ordered in time with later vertices to the left, and interleaved with free propagation $\mathrm{e}^{\mathcal{L}_S \Delta t}$ (or, in the interaction picture, absorbed into $\tilde{S}(t)$). Lines connect pairs of vertices:

- a *noise line* $S(t-s)$ connects $S^{\mathrm{q}}(t)$ to $S^{\mathrm{q}}(s)$ (q–q on the system side is c–c on the bath side);
- a *response line* $-i\chi(t-s)$ connects the *later* $S^{\mathrm{q}}(t)$ to the *earlier* $S^{\mathrm{c}}(s)$;
- there is no c–c line on the system side, and no line whose later end is S^{c}. Consequently the latest vertex of every diagram is S^{q}, which preserves Tr_S (trace preservation of Π) and is the system-side image of $\langle \phi_{\mathrm{q}} \phi_{\mathrm{q}} \rangle = 0$.

Example: the second-order term and the Born kernel

From Eq. (9), with $\mathcal{L}_I = -i(S^{\mathrm{q}}B^{\mathrm{c}} + S^{\mathrm{c}}B^{\mathrm{q}})$,

$$\mathrm{Tr}_B\left[\tilde{\mathcal{L}}_I(t_1)\tilde{\mathcal{L}}_I(t_2)\,\rho_S\otimes\rho_B\right] = -\Big[\tilde{S}_1^{\mathrm{q}}\tilde{S}_2^{\mathrm{q}}\underbrace{\mathrm{Tr}_B[B_1^{\mathrm{c}}B_2^{\mathrm{c}}\rho_B]}_{S(t_1-t_2)} + \tilde{S}_1^{\mathrm{q}}\tilde{S}_2^{\mathrm{c}}\underbrace{\mathrm{Tr}_B[B_1^{\mathrm{c}}B_2^{\mathrm{q}}\rho_B]}_{-i\chi(t_1-t_2)} + \tilde{S}_1^{\mathrm{c}}\tilde{S}_2^{\mathrm{q}}\underbrace{\mathrm{Tr}_B[B_1^{\mathrm{q}}B_2^{\mathrm{c}}\rho_B]}_{0} + \tilde{S}_1^{\mathrm{c}}\tilde{S}_2^{\mathrm{c}}\underbrace{\mathrm{Tr}_B[B_1^{\mathrm{q}}B_2^{\mathrm{q}}\rho_B]}_{0}\Big]\rho_S, \tag{24}$$

with the shorthand $\tilde{S}_i^a = \tilde{S}^a(t_i)$, $B_i^a = \tilde{B}^a(t_i)$. Only two of the four index combinations survive, exactly as the rules say. Going back to the Schrödinger picture, the surviving bracket is $-S^{\mathrm{q}}\mathrm{e}^{\mathcal{L}_S\tau}[S(\tau)S^{\mathrm{q}} - i\chi(\tau)S^{\mathrm{c}}]$ with $\tau = t_1 - t_2$, which is the second-order NZ kernel $K^{(2)}(\tau)$ of Eq. (29) below. If one instead expands in the $\pm$ basis and uses the table of Section 2.3, one obtains the expression $-[S, C(\tau)S\rho - C(-\tau)\rho S]$ (four terms once the commutator is expanded) familiar from textbook Born–Redfield derivations [8]; the rotated form is the same thing with the noise and response parts kept apart.

Example: pure dephasing to fourth order, and the first irreducible diagram

Take $H_S = \frac{\omega_0}{2}\sigma_z$ and $H_I = \frac{1}{2}\sigma_z \otimes B$, so $S = \sigma_z/2$. On the coherence $X = |0\rangle\langle 1|$ the qubit example of Section 2.2 gives $S^{\mathrm{q}}X = X$ and $S^{\mathrm{c}}X = 0$; moreover $S^{\mathrm{c/q}}$ commute with $\mathcal{L}_S$, so the free evolution just contributes $\mathrm{e}^{-i\omega_0 t}$ overall. Every response line ends on an S^{c} and therefore vanishes on X. What remains is a sum over pairings of noise lines with all vertices equal to 1:

$$\tilde{\rho}_{01}(t) = \rho_{01}(0)\Big[1 - \int_{t_1>t_2} S(t_1-t_2) + \int_{t_1>t_2>t_3>t_4} \big[\underbrace{S_{12}S_{34}}_{\text{reducible}} + \underbrace{S_{13}S_{24}}_{\text{crossed}} + \underbrace{S_{14}S_{23}}_{\text{nested}}\big] - \dots\Big], \qquad S_{ij} \equiv S(t_i - t_j), \quad (25)$$

where the sign at order $2n$ is $(-i)^{2n} = (-1)^n$. The first fourth-order pairing can be cut between t_2 and t_3 without cutting a line: it is the reducible diagram of Fig. 1 and equals a product of two second-order pieces. The crossed and nested pairings cannot be cut and are the fourth-order contributions to the NZ kernel $K^{(4)}$. Summing all orders, the three fourth-order pairings together with their higher-order relatives reassemble into the exponential $\exp[-\int_{t_1>t_2} S(t_1-t_2)] = \mathrm{e}^{-\Gamma(t)}$ of Eq. (38), because here all vertices commute. This is the simplest instance of the general statement in Section 3: the full series is a geometric (Dyson) resummation of the reducible structure, with the irreducible pieces as the kernel.

2.5 The influence functional: Wick's theorem with a commutative system

Suppose the system superoperators at different times commuted. Then the sum over all pairings of $2n$ vertices would be $\frac{1}{n!}$ times the n-th power of the single contraction, and the whole Dyson series would exponentiate. Operators do not commute, but in a path integral they are replaced by c-number paths: on the forward branch $S \to s^+(t)$, on the backward branch $S \to s^-(t)$, and in rotated variables $s^{\mathrm{c}} = \frac{1}{2}(s^+ + s^-)$, $s^{\mathrm{q}} = s^+ - s^-$. Repeating Section 2.4 with these c-numbers, the vertex $-i(S^{\mathrm{q}}B^{\mathrm{c}} + S^{\mathrm{c}}B^{\mathrm{q}})$ becomes $-i(s^{\mathrm{q}}B^{\mathrm{c}} + s^{\mathrm{c}}B^{\mathrm{q}})$ and the sum over pairings exponentiates to

$$\Phi[s^{\mathrm{c}}, s^{\mathrm{q}}] = -\int_0^t \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2 \Big[s^{\mathrm{q}}(t_1)S(t_1-t_2)s^{\mathrm{q}}(t_2) - i\, s^{\mathrm{q}}(t_1)\chi(t_1-t_2)s^{\mathrm{c}}(t_2)\Big], \quad (26)$$

so that $\rho_S(t) = \int \mathcal{D}s^{\pm}\, \mathrm{e}^{iS_S[s^+]-iS_S[s^-]+\Phi[s^{\mathrm{c}},s^{\mathrm{q}}]}(\dots)$. This is the Feynman–Vernon influence functional in Keldysh-rotated form [6, 9, 10]: S is the noise kernel and χ the dissipation kernel. Equivalently, $\Phi = -\frac{1}{2}\iint s^a D^{ab} s^b$ with D the Larkin–Ovchinnikov matrix (21), which is how it appears when one integrates out a Gaussian bath in the Keldysh path integral [1]. For a single Gaussian bath, Eq. (26) contains *all* the information that any of the methods below will use; they differ only in how they organize the non-commutativity that the path integral pushes into the measure.

Example: reading the influence functional

(i) Pure dephasing. With $S = \sigma_z/2$ and the system restricted to the coherence, the path variables are $s^+ = \frac{1}{2}$, $s^- = -\frac{1}{2}$, so $s^{\mathrm{c}} = 0$ and $s^{\mathrm{q}} = 1$. Then $\Phi = -\int_{t_1>t_2} S(t_1 - t_2) = -\Gamma(t)$ and $\rho_{01}(t) = \rho_{01}(0)\mathrm{e}^{-i\omega_0 t - \Gamma(t)}$, reproducing the fourth-order example above to all orders in one line. *(ii) Langevin form.* Φ is at most quadratic in s^{q}. Decouple the quadratic (noise) term with an auxiliary real field ξ, $\exp[-\frac{1}{2}\iint s^{\mathrm{q}} S s^{\mathrm{q}}] = \int \mathcal{D}\xi \, \exp[-\frac{1}{2}\iint \xi S^{-1}\xi + i\int \xi s^{\mathrm{q}}]$, and expand the system action to first order in s^{q} (the semiclassical step; exact for a harmonic system). The s^{q} integral then produces a delta functional that enforces an equation of motion for s^{c} with damping kernel χ and a Gaussian force ξ of correlator S. The q field has become the Martin–Siggia–Rose response field, the qq $= 0$ rule is the statement that the response field has no self-correlation (causality), and in the high-temperature limit $S(\omega) \to 2T\chi''(\omega)/\omega$ is the classical FDT. Section 4 returns to this from Mori's side.

Two things that are easy to get wrong. (1) The θ function in χ is not an approximation; it comes from the trace identity and is the reason causality is built in. If you ever find a $\chi(t-s)$ with $s > t$ in a system-side diagram, an index has been swapped. (2) The $-i$ in the response line is a convention tied to $\chi = i\theta\langle[B,B]\rangle$. With Kamenev's $D^R = -i\theta\langle[\phi,\phi]\rangle$ the line is $+D^R$. Keep one convention and check it on the single-mode example above.

3 Nakajima–Zwanzig and TCL as Dyson equations

3.1 The NZ equation

Take $\mathcal{P}X = \rho_B \otimes \mathrm{Tr}_B X$ and $\mathcal{Q} = 1 - \mathcal{P}$ [8, 11, 12]. Our assumptions give $\mathcal{P}\mathcal{L}_I\mathcal{P} = 0$ and $[\mathcal{P}, \mathcal{L}_S + \mathcal{L}_B] = 0$. For a factorized initial state the inhomogeneous term vanishes, and

$$\dot{\rho}_S(t) = \mathcal{L}_S\rho_S(t) + \int_0^t \mathrm{d}s\, K(t-s)\,\rho_S(s), \qquad K(\tau)X = \mathrm{Tr}_B\big[\mathcal{L}_I\,\mathrm{e}^{\mathcal{Q}\mathcal{L}\tau}\,\mathcal{Q}\mathcal{L}_I\,(X\otimes\rho_B)\big]. \tag{27}$$

In Laplace space, $\hat{f}(z) = \int_0^\infty \mathrm{d}t\,\mathrm{e}^{-zt}f(t)$ and

$$\boxed{\hat{\Pi}(z) = \frac{1}{z - \mathcal{L}_S - \hat{K}(z)} \quad\Longleftrightarrow\quad \hat{\Pi} = \hat{\Pi}_0 + \hat{\Pi}_0\,\hat{K}\,\hat{\Pi}, \qquad \hat{\Pi}_0 = (z-\mathcal{L}_S)^{-1}.} \tag{28}$$

3.2 Why K is the irreducible self-energy

Insert the Wick expansion of Section 2 into $\Pi(t)$. Every contribution is a time-ordered string of system superoperators, dressed by contraction lines and interleaved with $\mathrm{e}^{\mathcal{L}_S\Delta t}$. Now cut the time axis between any two consecutive vertices. If no line crosses the cut, the bath returns to ρ_B there and the diagram factorizes as $\cdots\Pi_0\,\mathcal{P}\cdots$. The factor $\mathcal{Q}$ inside $\mathrm{e}^{\mathcal{Q}\mathcal{L}\tau}\mathcal{Q}$ in Eq. (27) removes exactly these returns to ρ_B. Hence

Identity

$K = \sum\{\text{irreducible real-time diagrams}\}$, and Eq. (28) is the Dyson equation of Keldysh real-time diagrammatics for the reduced density matrix [7, 13–16].

For fermionic reservoirs the same statement holds, with sign factors from line crossings; this is the Schoeller–Schön setting. For non-Gaussian baths, higher cumulants replace pairwise lines.

In both cases irreducibility is defined with respect to the *reference state* ρ_B that appears in $\mathcal{P}$. Changing the projector (for example, to correlated projectors) changes the kernel.

3.3 Second order: the Keldysh structure made explicit

At order $\mathcal{L}_I^2$ we can drop $\mathcal{Q}$ in Eq. (27) because $\mathcal{P}\mathcal{L}_I\mathcal{P} = 0$. Using Eqs. (4) and (22),

$$\boxed{K^{(2)}(\tau) = -\sum_{\alpha\beta} S_\alpha^{\mathrm{q}}\, \mathrm{e}^{\mathcal{L}_S\tau}\Big[S_{\alpha\beta}(\tau)\, S_\beta^{\mathrm{q}} - i\,\chi_{\alpha\beta}(\tau)\, S_\beta^{\mathrm{c}}\Big].} \tag{29}$$

Written out, this is the familiar Born kernel

$$K^{(2)}(\tau)X = -\sum_{\alpha\beta}\big[S_\alpha, \mathrm{e}^{\mathcal{L}_S\tau}\big(C_{\alpha\beta}(\tau)S_\beta X - C_{\beta\alpha}(-\tau)XS_\beta\big)\big]. \tag{30}$$

The rotated form in Eq. (29) shows the two separate bath inputs, noise and response, and so connects directly to the influence functional (26).

Markov and Lindblad limits. Replace $\rho_S(s) \to \mathrm{e}^{-\mathcal{L}_S(t-s)}\rho_S(t)$ and extend the upper limit of integration to ∞; this gives Redfield's equation. The secular approximation then gives the Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) form, with rates set by $C(\omega) = S(\omega) + \chi''(\omega)$, which equals $2\chi''(\omega)[1 + n_B(\omega)]$ in equilibrium. In the field-theory language, a Lindblad equation corresponds to a Keldysh action with time-local Σ^R and frequency-independent Σ^K [17]:

$$S_K = \int \mathrm{d}t\Big[\bar\psi_+ i\partial_t\psi_+ - \bar\psi_- i\partial_t\psi_- - H_+ + H_- - i\sum_j\Big(L_{j+}\bar L_{j-} - \tfrac{1}{2}\big(\bar L_{j+}L_{j+} + \bar L_{j-}L_{j-}\big)\Big)\Big]. \tag{31}$$

3.4 TCL and the NZ–TCL map

The time-convolutionless (TCL) generator [18, 19] is $\mathcal{G}(t) = \dot\Pi(t)\Pi(t)^{-1}$, defined wherever $\Pi(t)$ is invertible. Substituting Eq. (27) gives

$$\mathcal{G}(t) = \mathcal{L}_S + \int_0^t \mathrm{d}s\, K(t-s)\,\Pi(s)\,\Pi(t)^{-1}, \tag{32}$$

an exact relation. Nestmann, Bruch and Wegewijs turn it into a fixed-point equation that generates $\mathcal{G}$ from K (and conversely) [20]. Reimer *et al.* compare five exact formulations, with attention to complete positivity and divisibility [21]. Three facts are worth keeping in mind:

- Truncating K and truncating $\mathcal{G}$ at the same order are *different* approximations. They agree at second order and differ from fourth order on [15, 16, 22].
- $\mathcal{G}(t)$ can be singular when $\Pi(t)$ is not invertible (for example, the resonant damped Jaynes–Cummings model at strong coupling [8]); K never is. "Local versus nonlocal" is therefore a question about the *representation*, not about the physics [23].
- With correlated initial states, the NZ inhomogeneity reappears. On the Keldysh side it corresponds to the imaginary-time (Matsubara) leg of the Kadanoff–Baym contour [24].

4 Mori's projection and the retarded self-energy

4.1 Generalized Langevin equation

Mori projects *operators*, not states [25–27]. Let $\dot{A} = i[H, A] \equiv i\mathsf{L}A$, choose an inner product (X, Y), and set $\mathsf{P}X = A\,(A,A)^{-1}(A,X)$, $\mathsf{Q} = 1 - \mathsf{P}$. Then, exactly,

$$\dot{A}(t) = i\Omega\, A(t) - \int_0^t \mathrm{d}s\, M(t-s)\, A(s) + F(t), \qquad F(t) = \mathrm{e}^{i\mathsf{QL}t}\,\mathsf{Q}\,i\mathsf{L}A, \tag{33}$$

with $i\Omega = (A, i\mathsf{L}A)(A,A)^{-1}$ and memory $M(t) = (F, F(t))(A,A)^{-1}$ (Kubo's second FDT [28]). The random force satisfies $(A, F(t)) = 0$, so the correlation $C(t) = (A, A(t))$ obeys

$$\hat{C}(z) = \frac{1}{z - i\Omega + \hat{M}(z)}\, C(0), \tag{34}$$

which again has Dyson form.

4.2 Worked example: the resonant level

Take $H = \varepsilon c^\dagger c + \sum_k \varepsilon_k d_k^\dagger d_k + \sum_k (V_k c^\dagger d_k + \text{h.c.})$, $A = c$, and the anticommutator inner product $(X,Y) = \langle\{X^\dagger, Y\}\rangle$, so that $(c,c) = 1$. Then $C(t) = \langle\{c^\dagger, c(t)\}\rangle$ and $G^R(t) = -i\theta(t)C(t)$. One finds $i\Omega = -i\varepsilon$ and $F(0) = -i\sum_k V_k d_k$. The Q-projected dynamics evolves d_k freely, so $M(t) = \sum_k |V_k|^2 \mathrm{e}^{-i\varepsilon_k t}$. Continuing $z \to -i\omega + 0^+$ in Eq. (34),

$$G^R(\omega) = \frac{1}{\omega - \varepsilon - \Sigma^R(\omega)}, \qquad \boxed{\Sigma^R(\omega) = -i\,\hat{M}(-i\omega + 0^+) = \sum_k \frac{|V_k|^2}{\omega - \varepsilon_k + i0^+}.} \tag{35}$$

This is the exact hybridization self-energy. For interacting systems the same construction reproduces the Dyson equation: the static (Hartree–Fock) part of Σ sits in Ω and the dynamical part in M [29]. Mori's continued fraction, obtained by iterating the projection, is the recursion method for G^R.

4.3 What Mori does *not* give you, and where Keldysh comes in

Equation (34) determines response functions, that is, the retarded sector. In equilibrium, the FDT then fixes the Keldysh component, $G^K = (1 - 2f)(G^R - G^A)$. Out of equilibrium, occupations, currents and noise live in $G^<$ (equivalently G^K), which the retarded function alone does not determine. In steady state without bound states, $G^< = G^R \Sigma^< G^A$ [5, 30]. Three routes supply the missing information:

1. *Density-matrix projection* (NZ, Section 3), which evolves the one-time state.
2. *Kadanoff–Baym equations* [24, 31] for two-time $G^{\lessgtr}$, reduced to one-time kinetic equations by the generalized Kadanoff–Baym ansatz (GKBA) [32]:

$$G^{\lessgtr}(t,t') \approx i\big[G^R(t,t')\,G^{\lessgtr}(t',t') - G^{\lessgtr}(t,t)\,G^A(t,t')\big]. \tag{36}$$

   GKBA plays the role of the projection onto one-time quantities, here the one-body density matrix $\rho^{(1)}(t) = -iG^<(t,t)$.

3. *Time-dependent Mori projections* with nonstationary reference states [33]. These produce two-time memory kernels and are closer in spirit to Kadanoff–Baym.

4.4 Generalized Langevin equation ↔ Keldysh saddle point

For a linear bath coupled to a coordinate x (Caldeira–Leggett / Ford–Kac–Mazur [34, 35]), apply a Hubbard–Stratonovich decoupling to the qq noise term in Eq. (26) with a real field ξ. The equation of motion $\delta/\delta x^{\mathrm{q}} = 0$ is then the quantum Langevin equation

$$m\ddot{x}^{\mathrm{c}} + V'(x^{\mathrm{c}}) - \int_{-\infty}^{t} \mathrm{d}s\, \chi(t-s)\, x^{\mathrm{c}}(s) = \xi(t), \qquad \langle \xi(t)\xi(s)\rangle = S(t-s), \tag{37}$$

which is Mori's Eq. (33) for $A = x$, with $M \leftrightarrow \chi$ and $F \leftrightarrow \xi$. Kubo's second FDT and the equilibrium relation between Σ^K and Σ^R are the same statement [1, 6].

5 Dictionary

Keldysh / NEGF	**Nakajima–Zwanzig / TCL**	**Mori**
Contour branches ±; c/q rotation	Left/right superoperators; $A^{\mathrm{c}} = \frac{1}{2}\{A, \cdot\}$, $A^{\mathrm{q}} = [A, \cdot]$	—
D^R (bath response)	χ on q–c lines	damping kernel in M
D^K (bath noise)	S on q–q lines	random-force statistics $(F, F(t))$
Dyson: $G = G_0 + G_0 \Sigma G$	$\hat{\Pi} = \hat{\Pi}_0 + \hat{\Pi}_0 \hat{K} \hat{\Pi}$	$\hat{C} = [z - i\Omega + \hat{M}]^{-1} C(0)$
Irreducible self-energy	NZ kernel K (irreducible diagrams)	memory function M ($\propto \Sigma^R$ for $A = c$)
Single-particle space	Liouville space of ρ_S (dimension d^2)	span of chosen operators
Retarded *and* Keldysh sectors	full one-time state	retarded sector only (out of eq.)
Kadanoff–Baym → GKBA	NZ → TCL (one-time closure)	time-dependent projections
Matsubara leg (initial correlations)	inhomogeneous term	initial-value dependence of F
Saddle point + Hubbard–Stratonovich ⇒ Langevin	—	GLE (33)
Local Σ^R, constant Σ^K	GKSL / Lindblad	Markovian $M(t) \propto \delta(t)$

Each approach has its natural territory. Keldysh field theory suits many-body problems, renormalization group, large-N and loop expansions, and self-consistent schemes (e.g. open-system DMFT [36]). NZ/TCL suits small systems, where questions about complete positivity, divisibility and memory are posed directly on Π. Mori suits the choice of relevant slow variables and continued-fraction approximations. Exact numerical bridges exist: memory kernels can be extracted from projection-free inputs [37] or from exact impurity solvers [38, 39], and hierarchical equations of motion (HEOM) [40] and process tensors [41] give independent exact references.

6 A sensing-flavoured example: pure dephasing done three ways

Take a qubit sensor $H_S = \frac{\omega_0}{2}\sigma_z$ with $H_I = \frac{1}{2}\sigma_z \otimes B$, a Gaussian bath, and a factorized initial state. On the coherence $X = |0\rangle\langle 1|$, $S^{\mathrm{q}}X = X$ and $S^{\mathrm{c}}X = 0$ (for $S = \sigma_z/2$). By Eq. (29), every response line then vanishes and only noise lines survive. Summing all pairings gives the exact result

$$\rho_{01}(t) = \rho_{01}(0)\, \mathrm{e}^{-i\omega_0 t}\, \mathrm{e}^{-\Gamma(t)}, \qquad \Gamma(t) = \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} \frac{S(\omega)}{\omega^2} F(\omega, t), \tag{38}$$

with $F = 1 - \cos\omega t$ for free induction decay. For a pulse sequence with sign function $y(s) = \pm 1$, $F(\omega, t) = \frac{\omega^2}{2}\left|\int_0^t y(s)\mathrm{e}^{i\omega s}\mathrm{d}s\right|^2$, the filter function of dynamical decoupling and noise spectroscopy [42, 43]. (Normalizations differ between papers.) The three descriptions compare as follows:

- **Keldysh**: Eq. (38) is exp of the qq part of the influence functional (26). Only D^K enters, so for pure dephasing a sensor measures the symmetrized noise and is blind to χ.
- **TCL**: $\mathcal{G}_{01}(t) = -i\omega_0 - \dot{\Gamma}(t)$ with $\dot{\Gamma}(t) = \int_0^t S(\tau)\mathrm{d}\tau$. TCL2 is *exact* here.
- **NZ**: $\hat{K}_{01}(z) = z - 1/\hat{\tilde{\rho}}_{01}(z)$ in the rotating frame is an *infinite* series in the coupling. This is the cleanest demonstration that "order n" means different things in different representations.

For metrology this matters directly: the time dependence of $\Gamma(t)$ at short times ($\propto t^2$ versus $\propto t$) changes the achievable precision scaling under non-Markovian noise [44].

7 Self-study plan (about 8 weeks, 6–8 h/week)

Each stage has a reading core and a derivation to do on paper. The stages are ordered so that every bridge is crossed only after both of its ends are familiar.

Week	Read	Goal / do
1	Breuer–Petruccione Ch. 9 [8]; de Vega–Alonso (NZ/TCL sections) [45]; Nakajima, Zwanzig originals [11, 12]	Derive Eqs. (27) and (28). **Ex. 1.**
2	Zwanzig Ch. 8 [26]; Mori [25]; Kubo [28]; Forster (memory functions) [27]	Derive the GLE (33); second FDT. **Ex. 2.**
3–4	Kamenev (contour, Keldysh rotation, dissipative oscillator) [1]; Kamenev–Levchenko [2]; Rammer–Smith [5]; Stefanucci–van Leeuwen (contour, Langreth rules) [24]	Keldysh rotation; causality structure; Langreth rules; $G^< = G^R\Sigma^< G^A$. **Ex. 3.**
5	Feynman–Vernon [9]; Grabert–Schramm–Ingold [10]; Weiss [6]; Sieberer–Buchhold–Diehl (Lindblad ↔ Keldysh) [17]	Derive Eq. (26) and the Keldysh action of a Lindblad equation. **Ex. 4.**
6	Schoeller–Schön [13]; König *et al.* [14]; Schoeller review [7]; Leijnse–Wegewijs [15]; Timm [16]	The central bridge: prove $K = \sum$(irreducible diagrams). **Ex. 5.**
7	Timm (TCL to all orders) [22]; Reimer *et al.* [21]; Nestmann–Bruch–Wegewijs [20]; Lipavský *et al.* (GKBA) [32]; Shi–Geva [37]; Cohen–Rabani [38]	NZ ↔ TCL map (32); GKBA as one-time closure. **Ex. 6.**
8	Degen–Reinhard–Cappellaro [43]; Cywiński *et al.* [42]; Chin–Huelga–Plenio [44]	Capstone (Exercise 7); write up a one-page summary of your own.

Optional extensions. Gurvitz–Prager infinite-bias Markovianity [46]; Kidon–Wilner–Rabani exact TCL generator [39]; HEOM [40]; process tensors [41]; open-system DMFT [36]; Grabert's time-dependent projections [33].

Exercises

Exercise 1. *Starting from Eqs. (4) and (27), derive Eq. (29). Show that* $\mathrm{Tr}_S K^{(2)}(\tau)X = 0$ *for all* X *and explain this as "the last vertex is always* S^{q}*".*

Exercise 2. *Reproduce* $\Sigma^R = -i\hat{M}(-i\omega + 0^+)$ *for the resonant level. Then add an on-site Hubbard* U *for a spinful level and identify the Hartree term in* Ω*.*

Exercise 3. *For Caldeira–Leggett, derive the quantum Langevin equation both from Mori's projection onto* (x, p) *and from the Keldysh saddle point with Hubbard–Stratonovich noise. Match* $M \leftrightarrow \chi$

and $(F, F(t)) \leftrightarrow S$.

Exercise 4. *From the GKSL equation, write the Keldysh action and read off* Σ^R *and* Σ^K *for a damped cavity mode. Check that they follow from Eq. (29) in the Born–Markov–secular limit.*

Exercise 5. *Prove that* $\mathcal{P}\mathcal{L}_I(\mathcal{Q}\,\mathrm{e}^{\mathcal{Q}\mathcal{L}\tau}\mathcal{Q})\mathcal{L}_I\mathcal{P}$ *generates exactly the irreducible real-time diagrams. (Hint: expand* $\mathrm{e}^{\mathcal{Q}\mathcal{L}\tau}$ *in* $\mathcal{L}_I$ *and show that every insertion of* $\mathcal{P}$ *corresponds to a cut with no crossing line.) Verify at fourth order by drawing all diagrams.*

Exercise 6. *For the pure-dephasing model of Section 6, compute* $\hat{K}_{01}(z)$ *to fourth order and check Eq. (32). Then do the damped Jaynes–Cummings model and locate the singularity of* $\mathcal{G}(t)$.

Exercise 7 (Capstone)**.** *Resonant level between two wide-band leads* (Γ_L, Γ_R)*, prepared empty at* $t = 0$*. (a) Keldysh: compute* $n(t) = -iG^<(t,t)$ *exactly. (b) NZ: extract the exact memory kernel for* $n(t)$ *and show that it becomes time-local in the infinite-bias limit [46]. (c) TCL: build* $\mathcal{G}(t)$ *from Eq. (32) and compare with its perturbative truncations. (d) Sensing: replace the leads by a Gaussian spin bath coupled via* σ_z *and relate your kernel to the filter-function formula (38) under a Hahn echo.*

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